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# Session: Presentation of Select Technical Papers

# Implementation of Resource Adequacy for Reliable RE Integration

***Presented By***

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# NEED FOR RESOURCE ADEQUACY

- India has set ambitious clean energy targets, known as five nectar elements or panchamrit:

50% of installed capacity from non-fossil fuel sources by 2030 → 500 GW non-fossil capacity by 2030 → Reduction of CO2 intensity of GDP by 45% from 2005 level → Creation of additional CO2 sink of 2.5-3 BT by 2030 → Net Zero by 2070

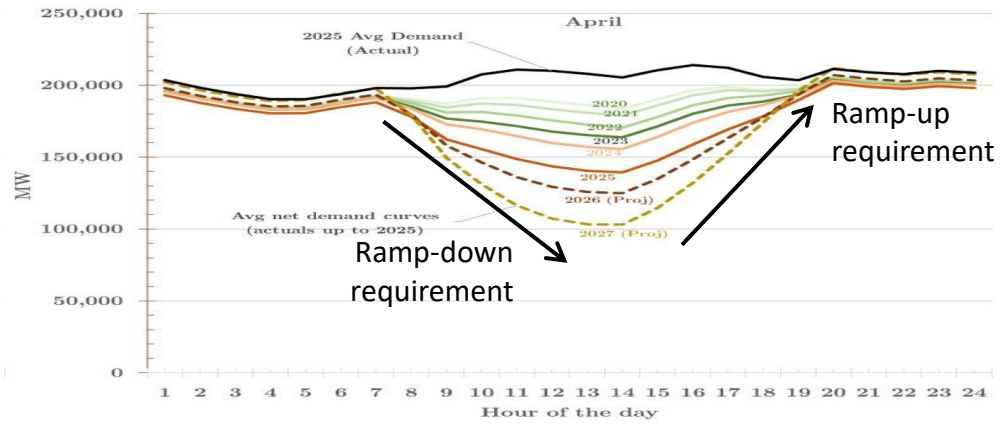
- As India embarks on this energy transition, its electricity grid faces several challenges, such as:

Intermittency at multiple levels

Unavailability during peak demand periods

Increased ramping requirement

Creation of duck curve situations




- Mitigating these challenges requires a paradigm shift in current planning processes towards Resource Adequacy (RA) which involves the planning of generation and transmission resources to reliably meet the projected demand in compliance with specified reliability standards with an optimal generation mix

# FOUR PILLARS OF RA FRAMEWORK

## Resource Adequacy Framework

### 1 Demand Assessment and Forecast

**1.1 Demand Forecast (ST/MT/LT)**

Forecast peak (MW) and energy requirement (MUs) over ST (1y), MT (5y), and LT (10y) at hourly granularity

Comprehensive inputs and drivers

Application of scientific and mathematical tools

### 2 Generation Resource Planning

**2.1 Capacity Crediting**

Realistic availability of resources during peak and critical hours

**2.2 Planning Reserve Margin**

Percent of projected resources above peak demand to capture contingencies and uncertainties

**2.3 RA Requirement and Allocation**

Incremental capacity required to meet peak + PRM considering supply discounted for CC

Allocation of RAR to states and discoms

### 3 Procurement Planning

**3.1 Procurement Resource Mix**

Identification of optimal and least-cost resource mix to meet RAR using energy modelling tools

**3.2 Procurement Type and Tenure**

Qualification of RAR and allocation into ST/MT/LT requirement to assist necessary procurement

**3.3 Capacity Trading/Sharing Construct**

Optimal procurement via various mechanism such as bilateral contracts, auctions, capacity markets etc.

### 4 Monitoring and Compliance

**4.1 Compliance Monitoring**

Ensure RA requirement is met on a continuous basis and the system operates in a reliable manner.

# NEED FOR STUDY

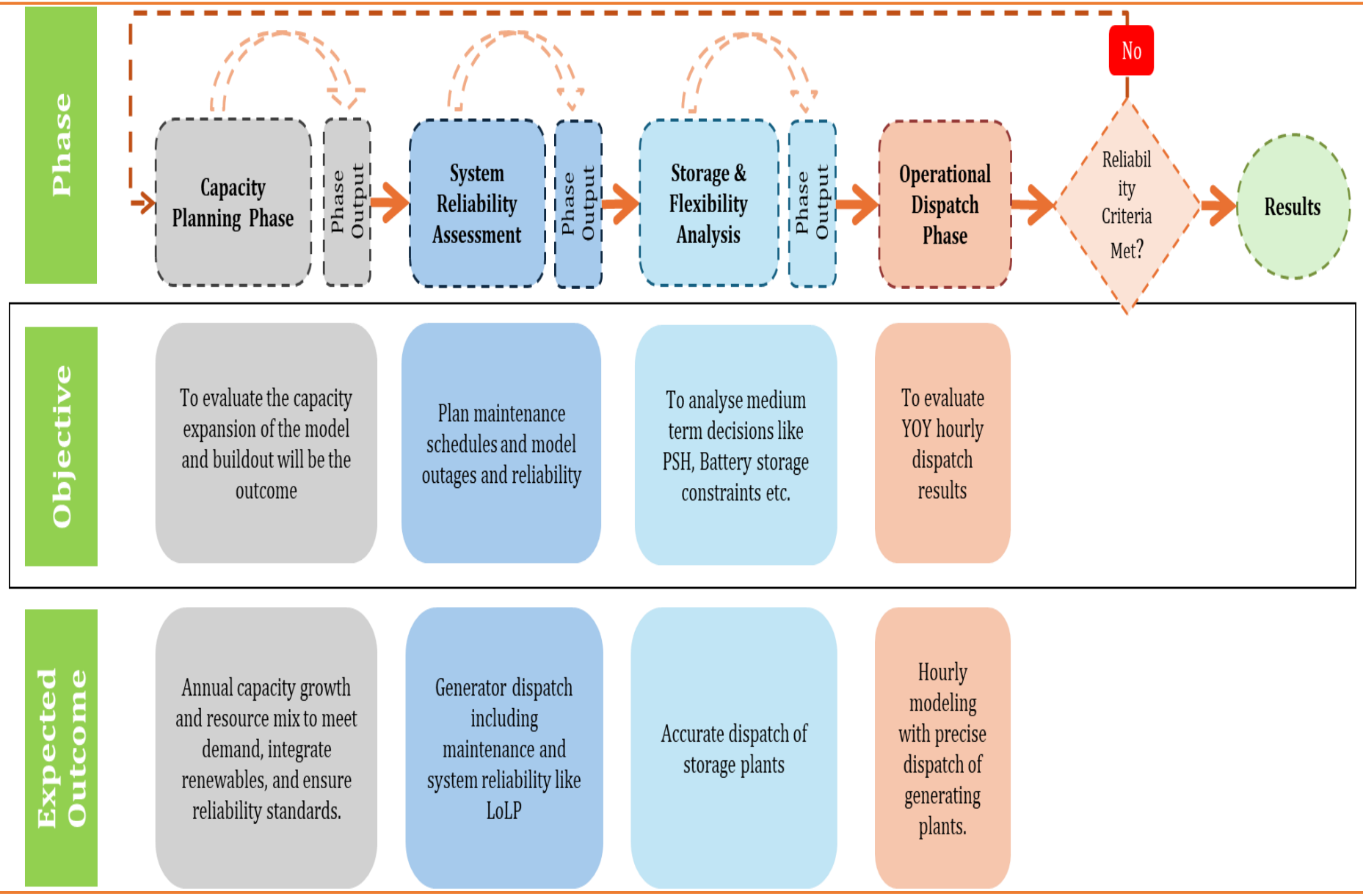
- **Mandate:** by MoP to SERCs (in Electricity (Amendment) Rules, 2022) to notify and implement RA Regulations
- **Status:** ~15 states have notified Final RA Regulations while ~3 have published Draft; some major RE-rich states like Rajasthan, Gujarat, and AP are yet to publish even Draft → entities are facing implementation hurdles and there is need for extensive capacity building and handholding
- **Critical aspect:** planning is driven by energy modelling and optimization techniques involving capacity expansion, dispatch optimization, and reliability assessment for meeting reliability standards set by CEA:

**Loss of Load Probability (LOLP)**
<= 0.2%

**Normalized Energy Not Served (NENS)**
<= 0.05%

- **Study:** Idam Infra has carried out a detailed RA modelling exercise for Maharashtra, Karnataka, and Rajasthan to identify an optimal resource mix adhering to reliability standards through FY34 → it serves as a scalable, sustainable model for reliable RE integration and RA planning for other states

# MODELLING PHILOSOPHY



# KEY INPUTS AND OUTPUTS

## Inputs

**1 Demand**
- Hourly profile for base year (FY24)
- Annual peak/energy projections for modelling horizon (FY25-FY34)

**2 Existing/Upcoming Resources: Technical**
- Capacity, heat rates, CUFs, ramp rates, start-up/shut-down times, maintenance, forced outages etc.
- Hourly profiles for RE

**3 Existing/Upcoming Resources: Financial**
- Capital costs, FOM, VOM, start-up/shut-down costs etc.
- Escalation factors

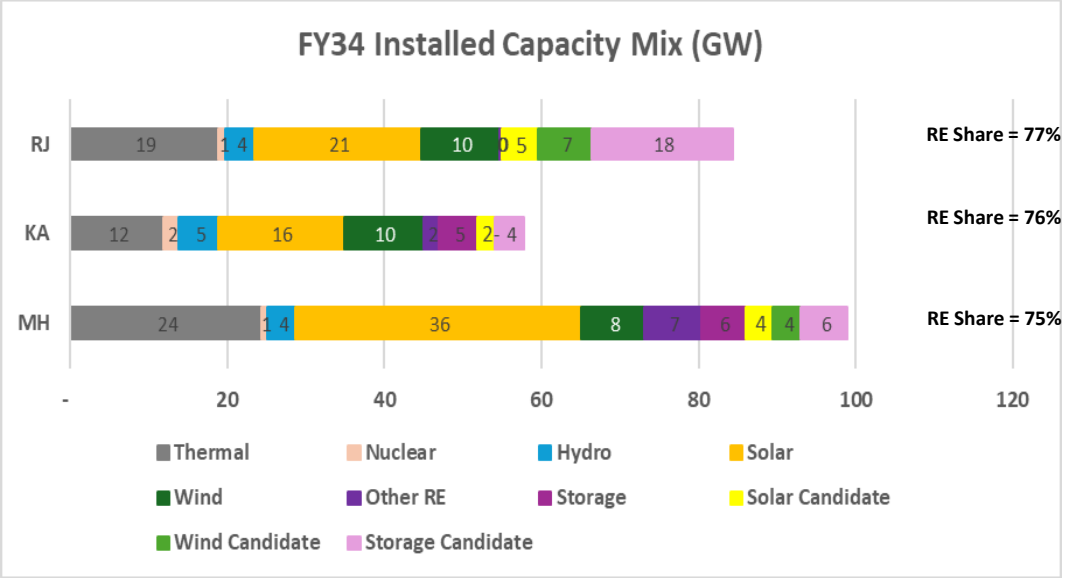
**4 Candidate Resources**
- Solar, Wind, BESS
- Technical /financial parameters

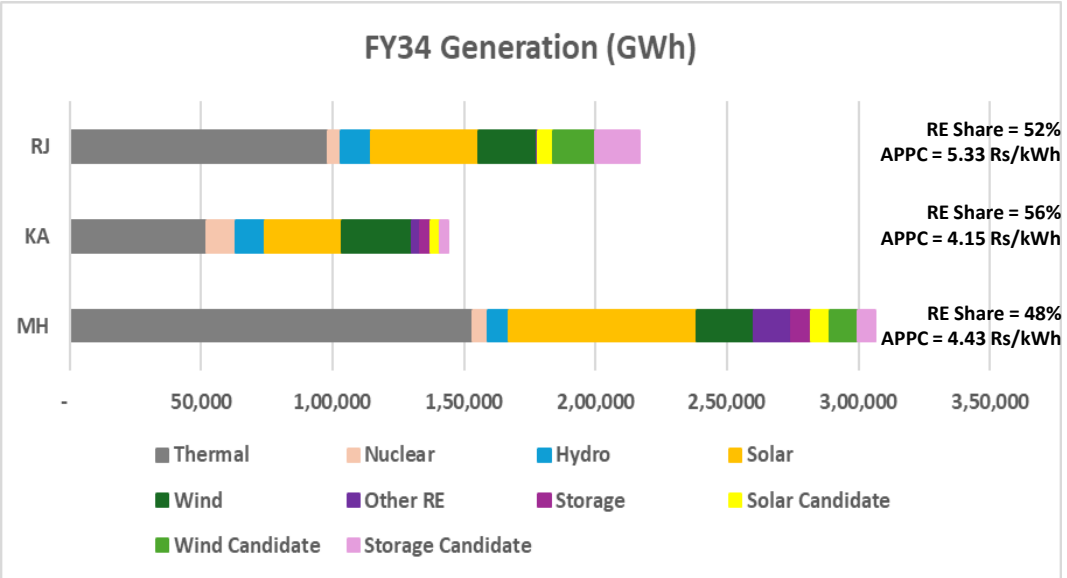
**5 Stochastics**
- **For hourly demand and RE profiles**
- Variance
- Standard deviation
- CV

## Outputs

- YoY resource mix
- Hourly load-generation balance
- Reliability metrics
- Generation costs

# RESULTS AND INSIGHTS




- In RJ and MH, future demand is met predominantly by solar + BESS
- Despite lesser demand, BESS buildout in RJ is ~3-5x times in MH/KA → because RJ has high demand requirement even in off-peak hours
- In KA, solar is complemented by existing wind leading to lesser BESS
- With only existing and contracted resources, RJ and MH violate while KA can satisfy reliability metrics of LOLP and NENS from FY25 → KA needs least capacity expansion
- With expansion, LOLP and NENS of all 3 states for all years are within limits, with PRM of ~15-25% by FY34
- Without new coal addition, demand and RA criteria are met by RE
- All 3 states are RPO compliant by FY30 & beyond
- All 3 states demonstrate cost-effective and reliable RE integration and RA planning for FY25-FY34

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# *THANK YOU*

For discussions/suggestions/queries email: isuw@isuw.in

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References: India's Panchamrit; Duck Curve; MoP Electricity Amendment Rules; MoP RA Guidelines; FoR Model Regulations